

OceanCove Restaurant Case

Instructor: Sairam Sriraman

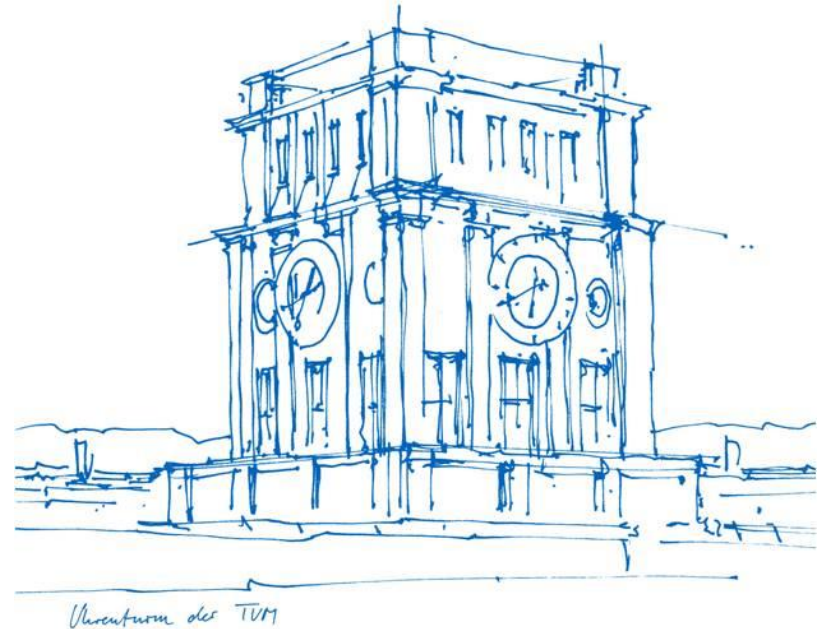
Examiner: Prof. Dr. David Wuttke

Technische Universität München

School of Management

Campus Heilbronn

Heilbronn, 2026



Case: OceanCove Restaurant

Capacity

Strategic aspects

Process analysis

Bottleneck

Service operations

Queuing



Consumer perspective

Operations performance

Case related questions

1. Describe OceanCove's target market, its needs, and values. How do the company's internal goals and operational activities allow it to meet these needs?
2. Draw a process flow diagram of a customer's experience at OceanCove from entering the restaurant to paying the bill
3. Draw a diagram of the kitchen operations. Display important operational information (e.g., process times, activity capacity, average queues, etc.)
4. At what rate do customers flow into the dining area at the peak of lunch hour? At what rate do order tickets flow into the kitchen at the peak of lunch hour?
5. Calculate the capacity per hour and capacity utilization at lunch time for each of the steps in the process. Where is the bottleneck at lunch time? What is the total restaurant capacity at lunch time?
6. How long would a customer have to wait for a grilled fish meal, assuming she is the first customer to enter the restaurant at lunch (or: to receive a rushed order)?
7. What if the customer arrives at the peak of lunch hour and her order is not rushed?
8. What are the implications for kitchen operations if the number of seats in the restaurant increases from 120 to 160? Would you recommend this? Should OceanCove open a new restaurant?

Case related questions

1. Describe OceanCove's target market, its needs, and values. How do the company's internal goals and operational activities allow it to meet these needs?
2. Draw a process flow diagram of a customer's experience at OceanCove from entering the restaurant to paying the bill
3. Draw a diagram of the kitchen operations. Display important operational information (e.g., process times, activity capacity, average queues, etc.)
4. At what rate do customers flow into the dining area at the peak of lunch hour? At what rate do order tickets flow into the kitchen at the peak of lunch hour?
5. Calculate the capacity per hour and capacity utilization at lunch time for each of the steps in the process. Where is the bottleneck at lunch time? What is the total restaurant capacity at lunch time?
6. How long would a customer have to wait for a grilled fish meal, assuming she is the first customer to enter the restaurant at lunch (or: to receive a rushed order)?
7. What if the customer arrives at the peak of lunch hour and her order is not rushed?
8. What are the implications for kitchen operations if the number of seats in the restaurant increases from 120 to 160? Would you recommend this? Should OceanCove open a new restaurant?

1. Value map: OceanCove Restaurants

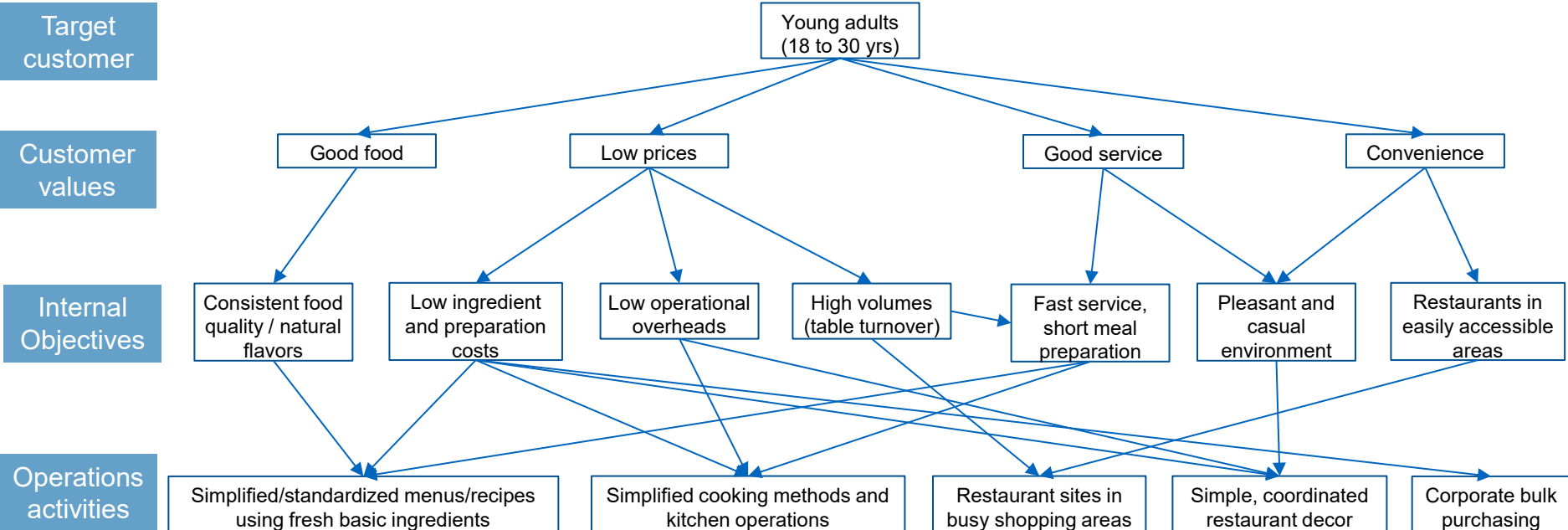
Target
customer

Customer
values

Internal
Objectives

Operations
activities

1. Value map: OceanCove Restaurants

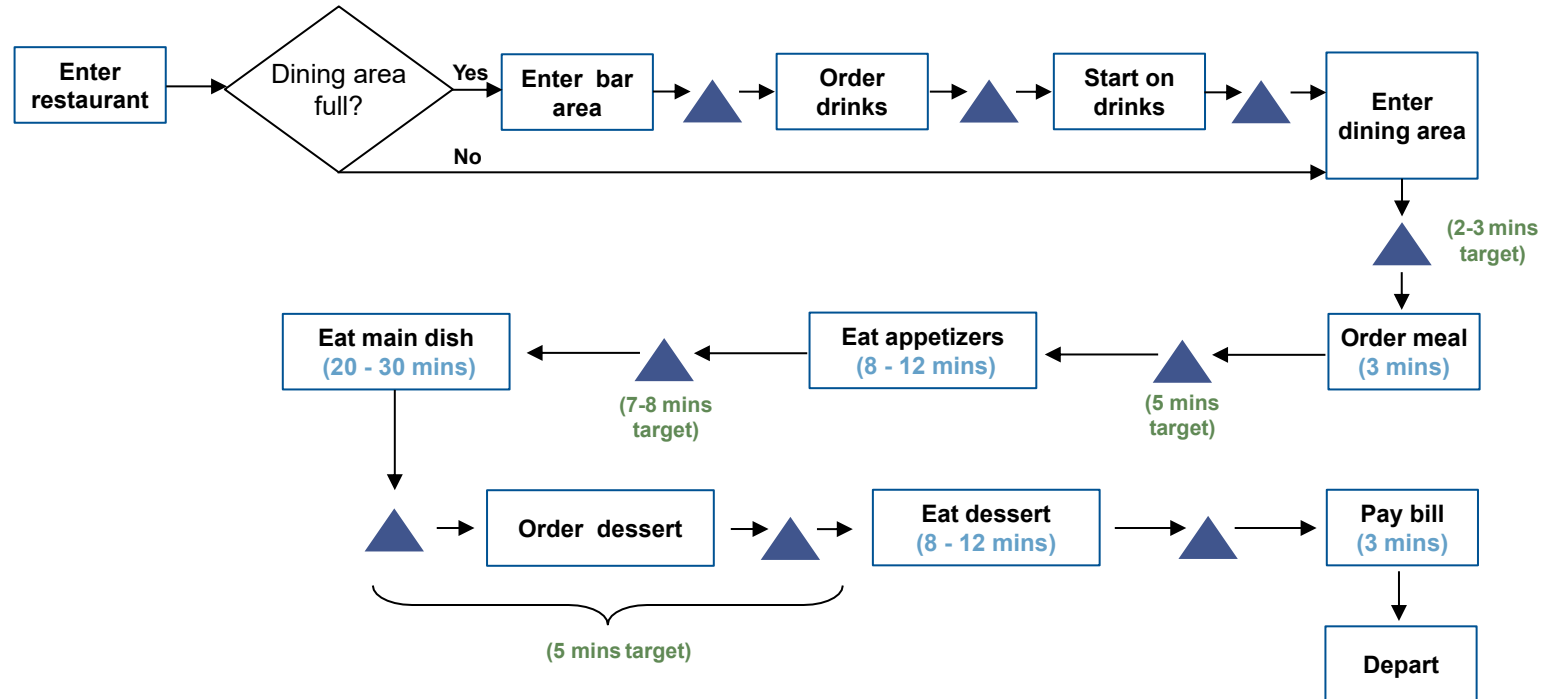


Case related questions

1. Describe OceanCove's target market, its needs, and values. How do the company's internal goals and operational activities allow it to meet these needs?
2. Draw a process flow diagram of a customer's experience at OceanCove from entering the restaurant to paying the bill
3. Draw a diagram of the kitchen operations. Display important operational information (e.g., process times, activity capacity, average queues, etc.)
4. At what rate do customers flow into the dining area at the peak of lunch hour? At what rate do order tickets flow into the kitchen at the peak of lunch hour?
5. Calculate the capacity per hour and capacity utilization at lunch time for each of the steps in the process. Where is the bottleneck at lunch time? What is the total restaurant capacity at lunch time?
6. How long would a customer have to wait for a grilled fish meal, assuming she is the first customer to enter the restaurant at lunch (or: to receive a rushed order)?
7. What if the customer arrives at the peak of lunch hour and her order is not rushed?
8. What are the implications for kitchen operations if the number of seats in the restaurant increases from 120 to 160? Would you recommend this? Should OceanCove open a new restaurant?

2. Process Flow Diagram for a Customer's Dining Experience during Lunch

2. Process Flow Diagram for a Customer's Dining Experience during Lunch

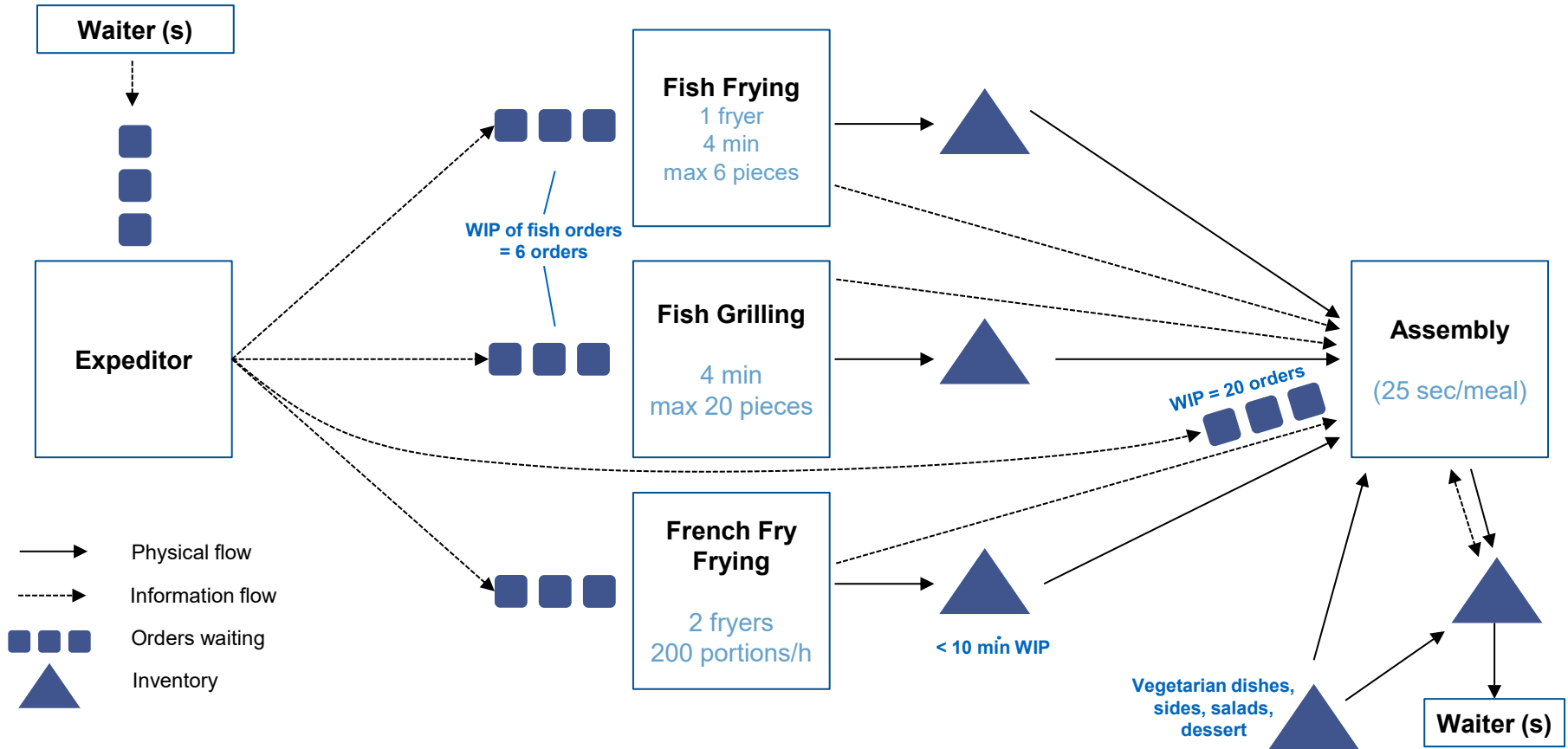


Case related questions

1. Describe OceanCove's target market, its needs, and values. How do the company's internal goals and operational activities allow it to meet these needs?
2. Draw a process flow diagram of a customer's experience at OceanCove from entering the restaurant to paying the bill
3. Draw a diagram of the kitchen operations. Display important operational information (e.g., process times, activity capacity, average queues, etc.)
4. At what rate do customers flow into the dining area at the peak of lunch hour? At what rate do order tickets flow into the kitchen at the peak of lunch hour?
5. Calculate the capacity per hour and capacity utilization at lunch time for each of the steps in the process. Where is the bottleneck at lunch time? What is the total restaurant capacity at lunch time?
6. How long would a customer have to wait for a grilled fish meal, assuming she is the first customer to enter the restaurant at lunch (or: to receive a rushed order)?
7. What if the customer arrives at the peak of lunch hour and her order is not rushed?
8. What are the implications for kitchen operations if the number of seats in the restaurant increases from 120 to 160? Would you recommend this? Should OceanCove open a new restaurant?

3. Process Flow Diagram for Kitchen Operations

3. Process Flow Diagram for Kitchen Operations



Case related questions

1. Describe OceanCove's target market, its needs, and values. How do the company's internal goals and operational activities allow it to meet these needs?
2. Draw a process flow diagram of a customer's experience at OceanCove from entering the restaurant to paying the bill
3. Draw a diagram of the kitchen operations. Display important operational information (e.g., process times, activity capacity, average queues, etc.)
4. At what rate do customers flow into the dining area at the peak of lunch hour? At what rate do order tickets flow into the kitchen at the peak of lunch hour?
5. Calculate the capacity per hour and capacity utilization at lunch time for each of the steps in the process. Where is the bottleneck at lunch time? What is the total restaurant capacity at lunch time?
6. How long would a customer have to wait for a grilled fish meal, assuming she is the first customer to enter the restaurant at lunch (or: to receive a rushed order)?
7. What if the customer arrives at the peak of lunch hour and her order is not rushed?
8. What are the implications for kitchen operations if the number of seats in the restaurant increases from 120 to 160? Would you recommend this? Should OceanCove open a new restaurant?

3. Demand: customer flow rate during peak lunch hour

What do we know?

Little's Law

How long are units in the process?

- **Average flow time**

How many units are in the process?

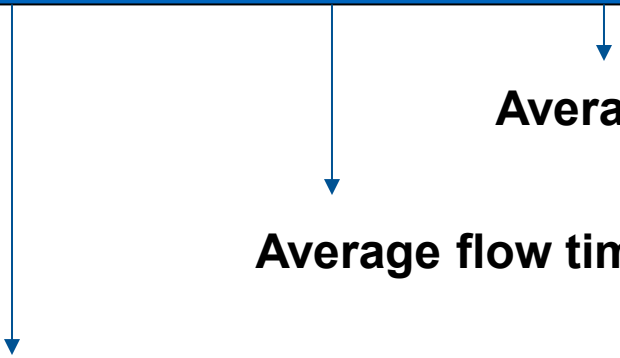
- **Average inventory**

How many units leave the process per hour?

- **Average flow rate**

$$\text{Average inventory} = (\text{Average flow time}) \times (\text{Average flow rate})$$

3. Demand: customer flow rate during peak lunch hour

$$\text{Average inventory} = (\text{Average flow time}) \times (\text{Average flow rate})$$


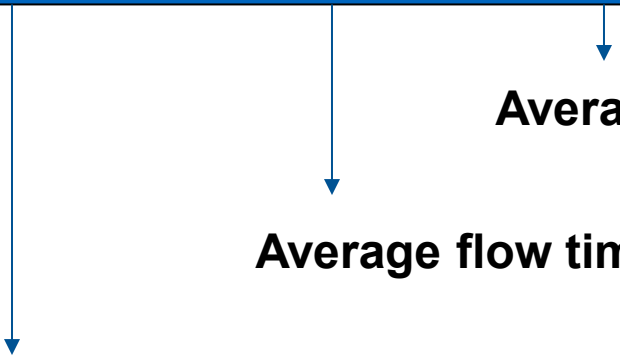
```
graph TD; A["Average inventory = (Average flow time) x (Average flow rate)"] --> B["Average flow rate = ?"]; A --> C["Average flow time = ?"]; A --> D["Average inventory = ?"];
```

Average flow rate = ?

Average flow time = ?

Average inventory = ?

3. Demand: customer flow rate during peak lunch hour

$$\text{Average inventory} = (\text{Average flow time}) \times (\text{Average flow rate})$$


Average flow rate = ?

Average flow time = 45 minutes per stay

Average inventory = 30 tables with an average of 3 customers on each

3. Demand: customer flow rate during peak lunch hour

Average inventory = (Average flow time) x (Average flow rate)

Average flow rate = 120 customers per hour

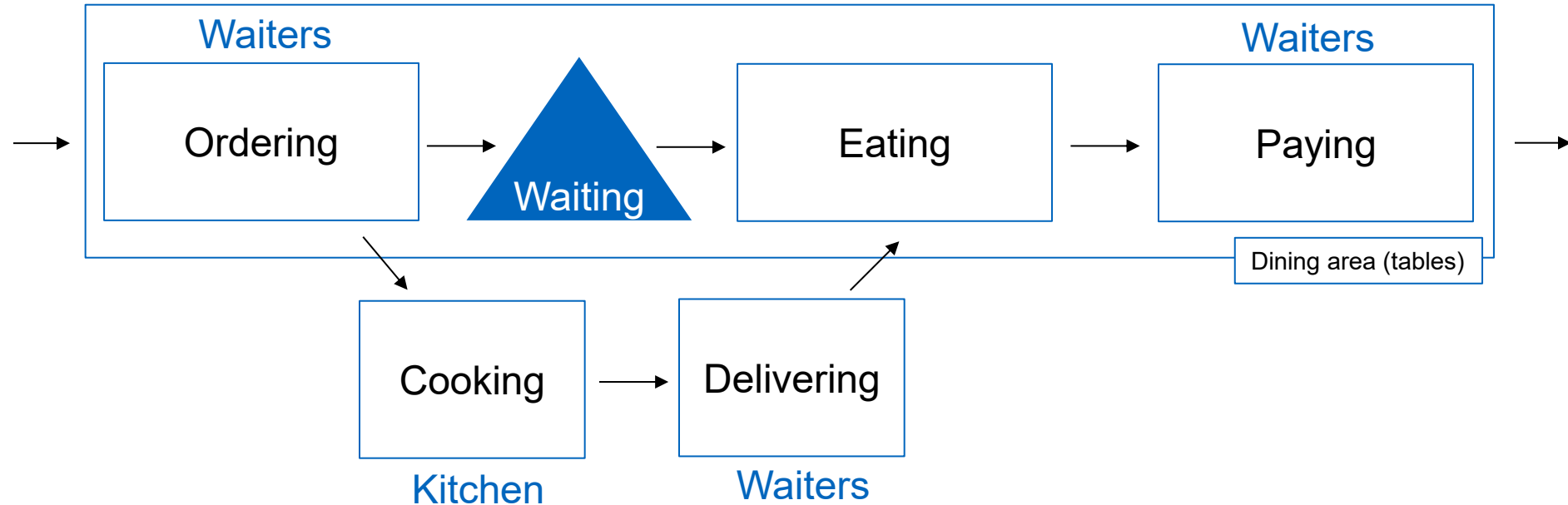
Average flow time = 45 minutes per stay

Average inventory = 30 tables with an average of 3 customers on each

Case related questions

1. Describe OceanCove's target market, its needs, and values. How do the company's internal goals and operational activities allow it to meet these needs?
2. Draw a process flow diagram of a customer's experience at OceanCove from entering the restaurant to paying the bill
3. Draw a diagram of the kitchen operations. Display important operational information (e.g., process times, activity capacity, average queues, etc.)
4. At what rate do customers flow into the dining area at the peak of lunch hour? At what rate do order tickets flow into the kitchen at the peak of lunch hour?
5. Calculate the capacity per hour and capacity utilization at lunch time for each of the steps in the process. Where is the bottleneck at lunch time? What is the total restaurant capacity at lunch time?
6. How long would a customer have to wait for a grilled fish meal, assuming she is the first customer to enter the restaurant at lunch (or: to receive a rushed order)?
7. What if the customer arrives at the peak of lunch hour and her order is not rushed?
8. What are the implications for kitchen operations if the number of seats in the restaurant increases from 120 to 160? Would you recommend this? Should OceanCove open a new restaurant?

5. Restaurant Capacity: Multiple Resources



$$\text{Restaurant capacity} = \min\{\text{kitchen capacity}, \text{waiter capacity}, \text{table capacity}\}$$

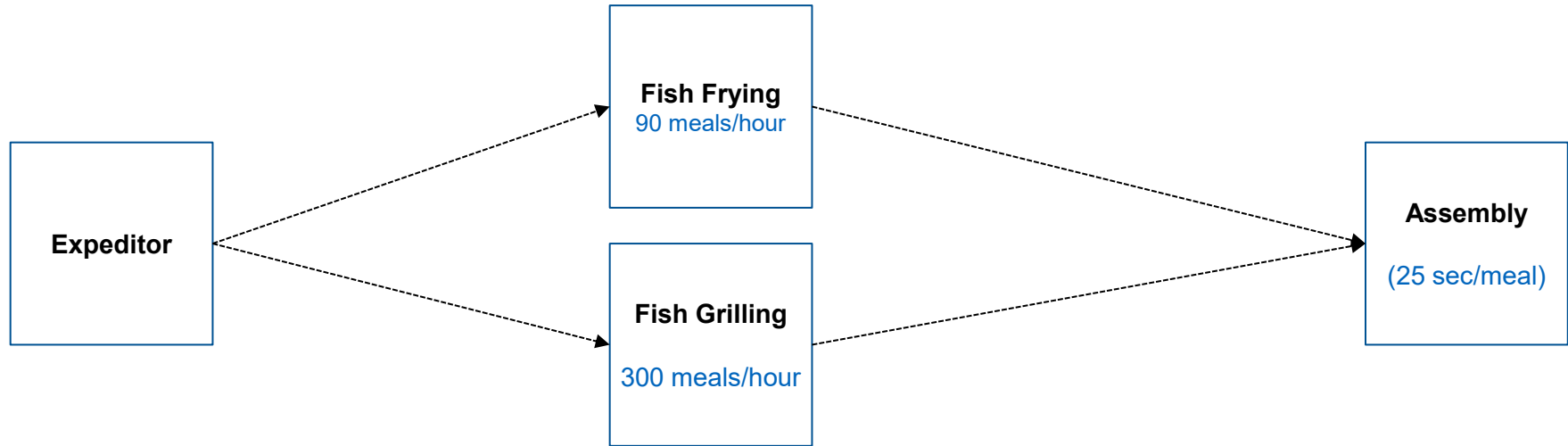
5. Restaurant capacity (lunch time)

Activity	Capacity (customers/hour)
Fish-Grilling	300 (= 20 portions x (60/4))
Fish-Frying*	90 (=6 pc. of fish / (4/60 hours per pc.)
French Fry-Frying	400 (= fryers x 200 servings/hour)
Expediting	plenty
Assembling	144 (= 1 chef/(0.42/60 hours per meal)
Order taking, Delivery and Billing	180 (= 6 waiters / (6/60 hours) = 60 tables/hour
Lunchtime Dining Area	160 (= 120 seats / (3/4 hour))

*) This analysis ignores the time for frying appetizers, which is probably small

5. Bottleneck analysis (lunch time)

“Grilled seafood tended to be about twice as popular as fried seafood.” (page 4)



Keeping the ratio 2:1, for 90 fried fish meals, there are 180 grilled fish meals, **totaling 270 meals/hour.**

5. Restaurant capacity (lunch time)

Activity	Capacity (customers/hour)
Fish menu (gilled or fried)	270 (= 90 + 180)
French Fry-Frying	400 (= fryers x 200 servings/hour)
Expediting	plenty
Assembling	144 (= 1 chef/(0.42/60 hours per meal)
Order taking, Delivery and Billing	180 (= 6 waiters / (6/60 hours) = 60 tables/hour
Lunchtime Dining Area	160 (= 120 seats / (3/4 hour))

*) This analysis ignores the time for frying appetizers, which is probably small

5. Restaurant capacity (dinner time)

Activity	Capacity (customers/hour)
Fish menu (gilled or fried)	270 (= 90 + 180)
French Fry-Frying	400 (= fryers x 200 servings/hour)
Expediting	plenty
Assembling	144 (= 1 chef/(0.42/60 hours per meal)
Order taking, Delivery and Billing	180 (= 6 waiters / (6/60 hours) = 60 tables/hour
Dinner Dining Area	87 (= 120 seats / (82.5/60 hour))

*) This analysis ignores the time for frying appetizers, which is probably small

Case related questions

1. Describe OceanCove's target market, its needs, and values. How do the company's internal goals and operational activities allow it to meet these needs?
2. Draw a process flow diagram of a customer's experience at OceanCove from entering the restaurant to paying the bill
3. Draw a diagram of the kitchen operations. Display important operational information (e.g., process times, activity capacity, average queues, etc.)
4. At what rate do customers flow into the dining area at the peak of lunch hour? At what rate do order tickets flow into the kitchen at the peak of lunch hour?
5. Calculate the capacity per hour and capacity utilization at lunch time for each of the steps in the process. Where is the bottleneck at lunch time? What is the total restaurant capacity at lunch time?
6. How long would a customer have to wait for a grilled fish meal, assuming she is the first customer to enter the restaurant at lunch (or: to receive a rushed order)?
7. What if the customer arrives at the peak of lunch hour and her order is not rushed?
8. What are the implications for kitchen operations if the number of seats in the restaurant increases from 120 to 160? Would you recommend this? Should OceanCove open a new restaurant?

6. Fastest possible lead time?

Activity	Duration
Order taking	3 mins
Grilling (assume precooked)	4 mins
Assembly	25 secs
Delivery	3 mins
Total	10 mins 25 sec

Case related questions

1. Describe OceanCove's target market, its needs, and values. How do the company's internal goals and operational activities allow it to meet these needs?
2. Draw a process flow diagram of a customer's experience at OceanCove from entering the restaurant to paying the bill
3. Draw a diagram of the kitchen operations. Display important operational information (e.g., process times, activity capacity, average queues, etc.)
4. At what rate do customers flow into the dining area at the peak of lunch hour? At what rate do order tickets flow into the kitchen at the peak of lunch hour?
5. Calculate the capacity per hour and capacity utilization at lunch time for each of the steps in the process. Where is the bottleneck at lunch time? What is the total restaurant capacity at lunch time?
6. How long would a customer have to wait for a grilled fish meal, assuming she is the first customer to enter the restaurant at lunch (or: to receive a rushed order)?
7. What if the customer arrives at the peak of lunch hour and her order is not rushed?
8. What are the implications for kitchen operations if the number of seats in the restaurant increases from 120 to 160? Would you recommend this? Should OceanCove open a new restaurant?

7. Average Flow Rate

Average inventory = (Average flow time) x (Average flow rate)

Average flow rate = 2 meals / minute

Average dining time = 45 minutes per stay

Average inventory = 30 tables with an average of 3 customers on each

Lead time during peak

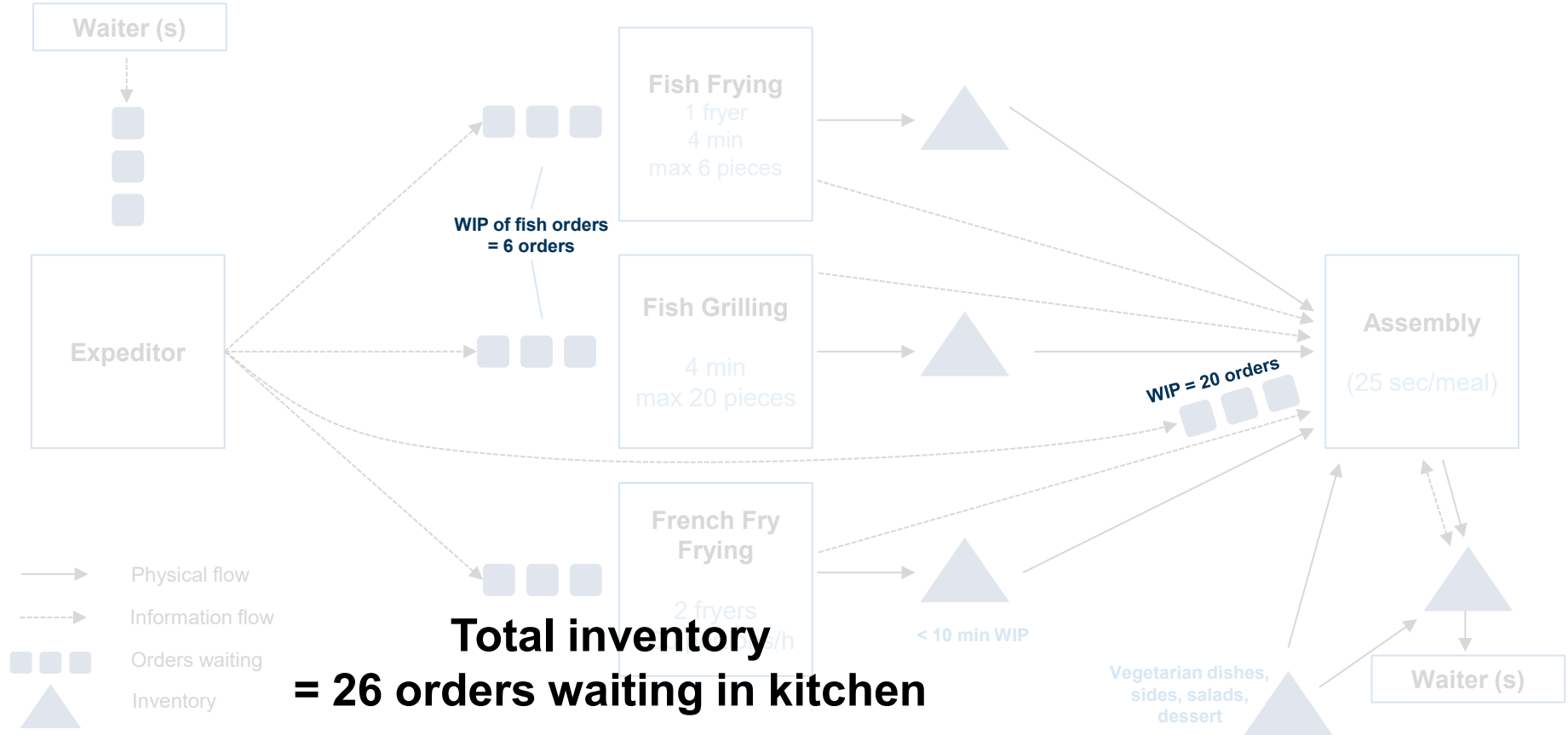
Average inventory = (Average flow time) x (Average flow rate)

Average flow rate = 2 meals / minute

Average waiting (!) time = ? minutes per meal

Average inventory = ? orders waiting in kitchen

7. Process Flow Diagram for Kitchen Operations



Lead time during peak

Average inventory = (Average flow time) x (Average flow rate)

Average flow rate = 2 meals / minute

Average waiting (!) time = 13 minutes per meal

Average inventory = 26 orders waiting in kitchen

7. Customer order lead time during peak

Average waiting (!) time = 13 minutes per meal

Fastest possible lead time (without wait) = 10 minutes 25 seconds per meal

Customer order lead time = 23 minutes 25 seconds per meal

Case related questions

1. Describe OceanCove's target market, its needs, and values. How do the company's internal goals and operational activities allow it to meet these needs?
2. Draw a process flow diagram of a customer's experience at OceanCove from entering the restaurant to paying the bill
3. Draw a diagram of the kitchen operations. Display important operational information (e.g., process times, activity capacity, average queues, etc.)
4. At what rate do customers flow into the dining area at the peak of lunch hour? At what rate do order tickets flow into the kitchen at the peak of lunch hour?
5. Calculate the capacity per hour and capacity utilization at lunch time for each of the steps in the process. Where is the bottleneck at lunch time? What is the total restaurant capacity at lunch time?
6. How long would a customer have to wait for a grilled fish meal, assuming she is the first customer to enter the restaurant at lunch (or: to receive a rushed order)?
7. What if the customer arrives at the peak of lunch hour and her order is not rushed?
8. What are the implications for kitchen operations if the number of seats in the restaurant increases from 120 to 160? Would you recommend this? Should OceanCove open a new restaurant?

8. Increase capacity to 160 seats?

BASE CASE WITH 120 SEATS

Lunch time capacity = 144 meals / hour

Lunch flow rate = 120 meals / hour (=160 customers / hour x $\frac{3}{4}$ seats per table used)

Lunch time revenues = \$6 x 120 meals / hour x 3 hours = **\$2,160**

Dinner time capacity = 87 meals / hour (= 120 / (82.5/60))

Dinner time flow rate = 65 meals / hour (=87 meals / hour x $\frac{3}{4}$ seats per table used)

Dinner time revenues = \$14 x 65 meals / hour x 4 hours = **\$3,640**

Daily revenues = **\$5,800**

8. Increase capacity to 160 seats?

NEW CASE WITH 160 SEATS

Lunch time capacity = 144 meals / hour (even though seating allows for 213)

Lunch flow rate = 144 meals / hour (=213 customers / hour x $\frac{3}{4}$ seats per table used)

Lunch time revenues = \$6 x 144 meals / hour x 3 hours = **\$2,592**

Dinner time capacity = 116 meals / hour (=160 / (82.5/60))

Dinner time flow rate = 87 meals / hour (=116 meals / hour x $\frac{3}{4}$ chairs per table used)

Dinner time revenues = \$14 x 87 meals / hour x 4 hours = **\$4,872**

Daily revenues = **\$7,464**

8. Increase capacity to 160 seats?

Assumption 1: Capacity utilization of 70%.

Assumption 2: Net profit margin 15%.

8. Increase capacity to 160 seats?

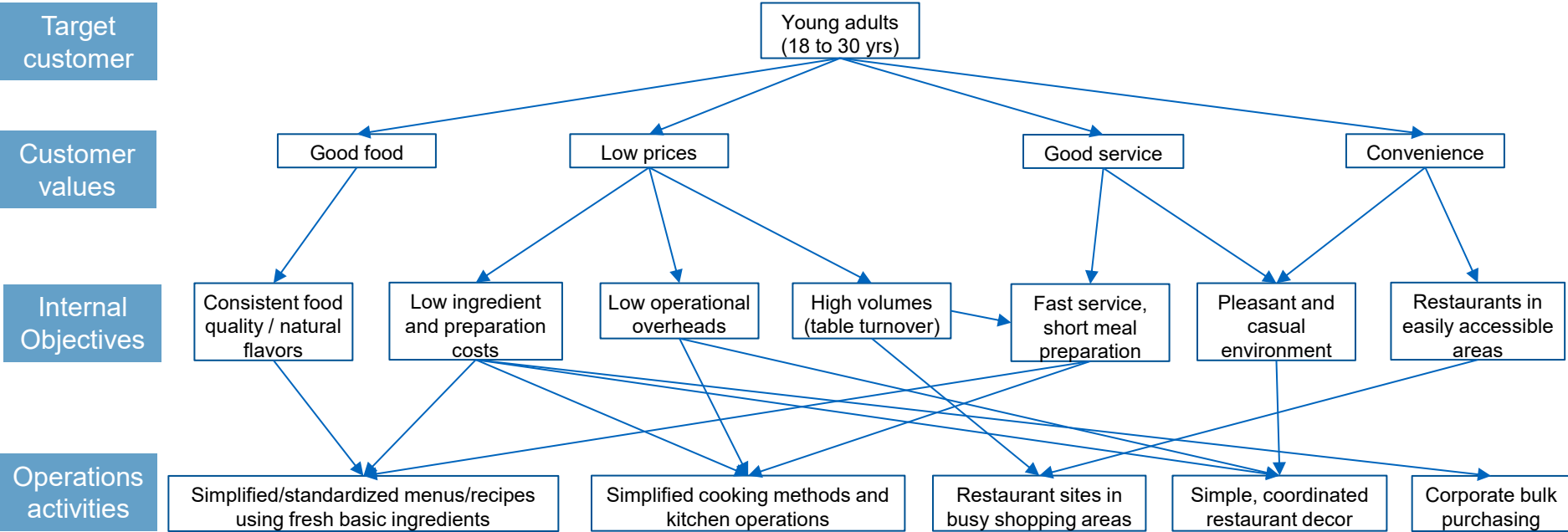
Assumption 1: Capacity utilization of 70%.

Assumption 2: Net profit margin 15%.

	120 seats	160 seats
Lunch time revenue (full cap util)	\$ 2.160	\$ 2.592
Lunch time revenue (at 70% cap util)	\$ 1.512	\$ 1.814
Lunch time contribution	\$ 227	\$ 272
Dinner time revenue (full cap util)	\$ 3.640	\$ 4.872
Dinner time revenue (at 70% cap util)	\$ 2.548	\$ 3.410
Dinner time contribution	\$ 382	\$ 512
Daily contribution	\$ 609	\$ 784

Further strategic implications

1. Value map: OceanCove Restaurants



Other thoughts?

